INCH-POUND

A-A-20100B
April 21, 1997
SUPERSEDING
A-A-20100A
September 5, 1986

COMMERCIAL ITEM DESCRIPTION

SHORTENING COMPOUNDS

The U.S. Department of Agriculture has authorized the use of this Commercial Item Description.

1. SCOPE.

1.1 This Commercial Item Description (CID) covers the following shortening compounds, packed in commercially acceptable containers, suitable for use by the Federal Government.

2. CLASSIFICATION.

2.1 The shortening compounds shall conform to the types in the following list which shall be specified in the solicitation, contract, or purchase order.

Types.

Type I	General purpose
Type II	Bakery (emulsifier-type)
Type III	Liquid shortening (deep-fry cooking)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, Defense Personnel Support Center, 2800 South 20th Street, ATTN: DPSC-HSL, Philadelphia, PA 19145-5099 or FAX (215) 737-2963.

FSC 8945

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3. SALIENT CHARACTERISTICS.

3.1 Type I: The general purpose shortening shall consist only of partially hydrogenated deodorized vegetable oil or blends of vegetable oils. The product shall not contain antioxidants or antifoaming agents. The shortening shall have a clean, fresh flavor and shall be free from rancidity, sourness, or other objectionable flavors or odors. The consistency shall be uniform and smooth. The texture shall be plastic and workable over a temperature range of 65E to 95EF (18.3E to 35EC).

3.2 Type II: The bakery (emulsifier-type) shortening shall contain vegetable fats and oils, which have been deodorized. The fats and oils shall have been processed by interesterification, hydrogenation, or both. The product may contain antioxidants. Antifoaming agents shall not be incorporated into the product. The shortening shall contain mono- and diglycerides in the amount specified in Table I. The product shall be white and have a clean, fresh flavor and shall be free from rancidity, sourness, or other objectionable flavors or odors. The consistency shall be uniform and smooth. The texture shall be plastic and workable over a temperature range of
65E to 95EF (18.3E to 35EC).

3.3 Type III: The liquid shortening (deep-fry cooking) shall consist only of deodorized vegetable fats and oils (excluding coconut oil, palm oil, and palm kernel oil), or blends of vegetable oils. The fats and oils shall have been processed by interesterification, hydrogenation, or both. The liquid shortening (deep-fry cooking) shall contain antifoaming agents and may contain antioxidants. The product shall have a clean, fresh flavor, free from rancidity, sourness, or other objectionable flavors or odors. The consistency shall be fluid at 70EF (21.1EC).

3.4 Type I, Type II, and Type III: The finished shortening products shall not have a saturated fatty acid content greater than 30 percent.

3.5 Analytical requirements: The finished products shall conform to the analytical requirements as specified in Table I and Table II. Analyses shall be made in accordance with Methods of the American Oil Chemists' Society (AOCS).

TABLE I. Analytical data

Requirements	Types I	II	III
Free fatty acids (percent), maximum	0.05	0.05	0.05

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TABLE I. Analytical data (Continued)

Requirements	Types I	II	III
Peroxide value (meq of peroxide per kg of fat), maximum	1.0	1.0	0.5
Wiley melting point (EF), range	112 to 120	110 to 120	-
Fat stability (AOM) (hours), minimum	75	40	25
Smoke point (EF), minimum	425	-	420
Alpha mono- and diglycerides, (percent), range	-	1.5 to 5.5	-
Moisture and volatile matter (percent), maximum	0.10	0.20 1/	0.10
Lovibond color (red), maximum	2.5	2.5	2.5

1/ Applicable to moisture only. If shortening is packed in cans, the maximum moisture shall not exceed 0.07 percent.

TABLE II. Solid fat index 1/

Type	Temperatures 50EF (10EC)	70EF (21.1EC)	92EF (33.3EC)
I	23 to 31	18 to 24	9 to 17
II	26 to 38	18 to 30	11 to 25
III	0 to 11	0 to 8	0 to 6

1/ As determined by the dilatometric measurement (AOCS Cd 10-57).

4. REGULATORY REQUIREMENTS.

4.1 The delivered shortening compounds shall comply with all applicable Federal and State mandatory requirements and regulations relating to the preparation, packaging, labeling, storage, distribution, and sale of the shortening compounds within the commercial

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marketplace. Delivered shortening compounds shall comply with all applicable provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The shortening compounds provided shall meet the salient characteristics of this CID, conform to the producer's own specifications, standards, and quality assurance practices, and be the same shortening compounds offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Quality assurance. When required in the solicitation, contract, or purchase order, the Federal Grain Inspection Service (FGIS), U.S. Department of Agriculture, shall determine the quality assurance of the shortening compounds according to FGIS procedures. The shortening compounds shall be examined or analyzed as applicable in accordance with applicable provisions in this CID, solicitation, contract, or purchase order, and, when applicable, the United States Standards for Condition of Food Containers currently in effect on the date of the solicitation.

5.3 Net weight. When required in the solicitation, contract, or purchase order, the net weight shall be as specified in Table III. The lot average net weight shall not be less than the specified net weight and no sample unit weight shall be less than the allowed unit net weight variation.

TABLE III. Net weights

	Average net weight and Unit net weight 1/	Container volume
Type I	3 pounds 5½ pounds 33 pounds 50-pound cube 2/	5 gallons
Type II	33 pounds 50-pound can or cube 2/	5 gallons
Type III	7 pounds, 10.5 ounces 9 pounds, 9 ounces 19 pounds, 2.5 ounces 400-pound, metal drum	4 quarts 5 quarts 10 quarts

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1/ Sample unit net weight variation allowed is minus 2 percent for containers with net weights up to 8 pounds and minus 1 percent for containers with net weights greater than 8 pounds.
2/ Polyethylene bag in fiberboard box.

6. PACKAGING.

6.1 Preservation, packaging, packing, labeling, and case marking. Preservation, packaging, packing, labeling, and case marking shall be as specified in the solicitation, contract, or purchase order.

7. NOTES.

7.1 Purchasers shall specify:

- Type(s) of shortening compounds required.

7.2 Sources of documents.

7.2.1 Source of information for nongovernmental document is as follows:

Copies of the Methods of the American Oil Chemists' Society may be obtained from: **American Oil Chemists' Society, P.O. Box 3489, 1608 Broadmoor Drive, Champaign, IL 61826-3489.**

7.2.2 Sources of information for governmental documents are as follows:

Applicable provisions of the Federal Food, Drug, and Cosmetic Act are contained in 21 CFR Parts 1 to 199. This document may be purchased from: **Superintendent of Documents, ATTN: New Orders, P.O. Box 371954, Pittsburgh, PA 15250-7954.** Credit card (MasterCard or Visa) purchases may be made by calling the Superintendent of Documents on (202) 512-1800.

Copies of the United States Standards for Condition of Food Containers are available from: **Chairperson, Condition of Container Committee, STOP 0243, 1400 Independence Avenue, SW, Washington, DC 20250-0243.**

Civil agencies and other interested parties may obtain copies of this CID from: **General Services Administration, Federal Supply Service, Specifications Section, Suite 8100, 470 East L'Enfant Plaza, SW, Washington, DC 20407.**

Military activities should submit requests for copies of this CID to: **Standardization**

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